

Topic to discuss

Proof of Euler's Method formula
in ODE

From calculus, if a function is sufficiently differentiable, then:

$$y(x_n+h) = y(x_n) + h y'(x_n) + \frac{h^2}{2!} y''(x_n) + \frac{h^3}{3!} y'''(x_n) + \dots$$

This is called Taylor Series Expansion.

It gives the exact value of the function near x .

Here, $y(x) \rightarrow$ present value

$h y'(x) \rightarrow$ linear change

$\frac{h^2}{2!} y''(x) \rightarrow$ curvature correction

and higher order terms.

Suppose the differential Equation is :

$$\frac{dy}{dx} = f(x, y)$$

Then, $y'(x) = f(x, y)$

Substitute into Taylor expansion,

$$y(x+h) = y(x) + hf(x, y) + \frac{h^2}{2!} y''(x) + \dots$$

Now in numerical notation,

$$x = x_n$$

$$y(x_n) = y_n$$

$$y(x_n + h) = y_{n+1}$$

$$\text{So, } y_{n+1} = y_n + hy'_n + \frac{h^2}{2!} y''_n + \frac{h^3}{3!} y'''_n + \dots$$

This is Taylor series Method.

It uses many derivatives terms for high accuracy.

To use this formula we must calculate,

$$y'', y''', y^{(4)}, \dots \text{ etc}$$

which becomes difficult for complicated equations.

So we try to simplify it.

Euler Method is obtained from Taylor series Method by keeping only the first two terms.

From Taylor method,

$$y_{n+1} = y_n + h y'_n + \frac{h^2}{2!} y''_n + \dots$$

Now neglect higher-order terms, $\frac{h^2}{2!} y''_n$ and so on

Then,
$$y_{n+1} \approx y_n + h y'_n$$

also we know,
$$y'_n = f(x_n, y_n)$$

Therefore,
$$y_{n+1} = y_n + h f(x_n, y_n)$$

That gives, new value = old value + (step size)(slope)

This is Euler's Method.

Why we Neglect Higher Terms

Because

calculation becomes simpler

Computation becomes faster

Only first derivative is needed.

But neglecting terms introduces error.

Euler method sacrifices accuracy for simplicity.

* for sufficiently small h , $\frac{h^2}{2}y_n''$, $\frac{h^3}{6}y_n''' \dots$ are very small, therefore

$$y_{n+1} \approx y_n + hy_n'$$

Error in Euler Method

Since we ignored, $\frac{h^2}{2!} y'' + \dots$

Actual solution, $y(x+h) = y(x) + hy'(x) + \frac{h^2}{2!} y''(x) + \dots$

Euler Approximation, $y_E = y(x) + hy'(x)$

Now we know,

$$\text{Error} = \text{Exact} - \text{Approximate}$$

$$\text{Error} = \frac{h^2}{2!} y'' + \frac{h^3}{3!} y''' + \dots$$

When h is small, $h^2 > h^3 > h^4 > \dots$

The leading term is: $\frac{h^2}{2} y''$

All remaining terms are even smaller.

Therefore,

$$\text{Local Error} = \frac{h^2}{2} y'' + O(h^3)$$

The dominant behaviour is $O(h^2)$

It does not mean $\text{Error} = h^2$

It means error behaves proportionally to h^2
when h becomes very small.

So, local truncation Error is $O(h^2)$

Number of steps, $N \approx \frac{1}{h}$

$$\therefore \text{Global Error} = \frac{1}{h} \times h^2 = h$$

$$\therefore \text{Global Error} = O(h)$$

Hence Euler's Method is obtained by retaining only the first two terms of the Taylor series expansion. It is a first-order numerical method with global error $O(h)$.

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